Criticism on GFS

Here is the summary of criticism on GFS's architecture.

Problems associated with single master

As GFS has grown in usage, Google has started to see the following problems with the centralized master scheme:

- Despite the separation of control flow (i.e., metadata operations) and data flow, the master is emerging as a bottleneck in the design. As the number of clients grows, a single master could not serve them because it does not have enough CPU power.
- Despite the reduced amount of metadata (because of the large chunk size), the amount of metadata stored by the master is increasing to a level where it is getting difficult to keep all the metadata in the main memory.

Problems associated with large chunk size

Large chunk size (64MB) in GFS has its disadvantages while reading. Since a small file will have one or a few chunks, the ChunkServers storing those chunks can become hotspots if a lot of clients are accessing the same file. As a workaround for this problem, GFS stores extra copies of small files for distributing the load to multiple ChunkServers. Furthermore, GFS adds a random delay in the start times of the applications accessing such files.